

DDE-SOLVER: A MAPLE PACKAGE FOR DISCRETE DIFFERENTIAL EQUATIONS

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ABSTRACT. We introduce [DDE-Solver](#), a Maple package designed for solving *Discrete Differential Equations* (DDEs). These equations are functional equations relating algebraically a formal power series $F(t, u)$ with polynomial coefficients in a “catalytic” variable u , with specializations of it with respect to the catalytic variable. Such equations appear in enumerative combinatorics, for instance in the enumeration of maps, trees, walks. Bousquet-Mélou and Jehanne showed in 2006 that when these equations are of a fixed point type in F , then F is an algebraic series. In the same paper, they proposed a systematic method for computing annihilating polynomials of these series. Bostan, Safey El Din and the author of this paper recently designed new efficient algorithms for computing these witnesses of algebraicity.

The current paper provides combinatorialists an automated tool in hand that solves DDEs using these algorithms. We compare the timings of [DDE-Solver](#) on DDEs from the literature.

1. INTRODUCTION

Sequences of non-negative integers are ubiquitous in enumerative combinatorics. For instance, when studying bicoloured maps (the faces are either coloured in purple or white, see Figure 6.2) such that the degree of each purple face is 3 and the degree of each white face is a multiple of 3, the introduction of the sequence $(c_n)_{n \in \mathbb{N}}$ of such maps, called *3-constellations*, with n purple faces starts with the numbers

$$1, 1, 6, 54, 594, 7371, 99144, \dots$$

For such a sequence, many questions arise: does there exist a finite way of representing the infinite amount of data given by this sequence? Does there exist a *closed formula* (i.e. an explicit formula) for $(c_n)_{n \in \mathbb{N}}$? What is the value of c_{n_0} , for $n_0 \in \mathbb{Z}_{\geq 0}$ large (e.g. $n_0 = 3 \cdot 10^6$), and how fast can it be computed? What is the asymptotic growth of c_n when $n \rightarrow \infty$?

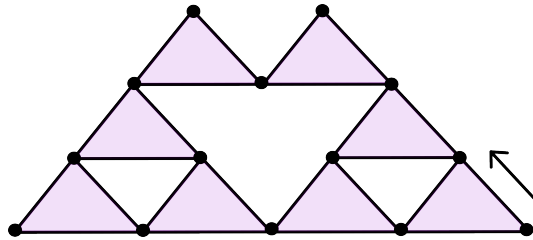


FIGURE 1. A 3-constellation contributing to the coefficient c_8 . Its white internal faces have either degree 3 or 6, while its external white face has degree 12.

In order to answer such questions, a common method is to introduce and study the properties of the associated generating function. For the above enumeration, one would introduce $G(t) := \sum_{n \in \mathbb{N}} c_n t^n \in \mathbb{Q}[[t]]$. For sophisticated enumerations, it is usually hard to study directly the generating series. Refining the initial counting problem usually leads to introducing a new variable – called *catalytic* in what follows –, which leads to considering a

bivariate generating function, say $F(t, u)$, where u is the catalytic variable. For well-chosen refinements, it is possible to write an equation over $\mathbb{Q}(t, u)$ relating $F(t, u)$ and $G(t)$.

For the above example of 3-constellations, one would typically refine the enumeration with the sequence $(c_{n,d})_{n,d \in \mathbb{N}}$, where $c_{n,d}$ is the number of 3-constellations having n purple faces and outer degree $3d$. Straightforwardly, one would introduce its associated generating function $F(t, u) = \sum_{n,d \geq 0} c_{n,d} u^d t^n \in \mathbb{Q}[u][[t]]$. As $\sum_{d=0}^{\infty} c_{n,d} = c_n$, it holds that $F(t, 1) = G(t)$. After analysis ([9, Fig. 7]) on the construction of these objects yields the equation

$$F(t, u) = 1 + tuF(t, u)^3 + tu(2F(t, u) + F(t, 1)) \frac{F(t, u) - F(t, 1)}{u - 1} + tu \frac{F(t, u) - F(t, 1) - (u - 1) \cdot \partial_u F(t, 1)}{(u - 1)^2}. \quad (1)$$

Let $k > 0$ and f, Q be polynomials. For $a \in \mathbb{Q}$, denote $\Delta_a : F \in \mathbb{Q}[u][[t]] \mapsto \frac{F(t, u) - F(t, a)}{u - a} \in \mathbb{Q}[u][[t]]$ and Δ_a^ℓ the ℓ^{th} iteration of the divided difference operator Δ_a . Then (1) is of the form

$$F(t, u) = f(u) + t \cdot Q(F(t, u), \Delta_a F(t, u), \dots, \Delta_a^k F(t, u), t, u). \quad (2)$$

Equations of the form (2) are called *discrete differential equations (DDEs)* of order k . For instance, (1) is a DDE of order $k = 2$, and the divided difference operator Δ_a is taken at the point $a = 1$. Observe that due to its fixed-point nature, (2) admits a unique solution in $\mathbb{Q}[u][[t]]$. Moreover, the following result proved by Bousquet-Mélou and Jehanne in [9], reminiscent of Popescu's theorem [23, Thm. 1.4], implies that this solution is annihilated by a nonzero polynomial with coefficients in $\mathbb{Q}(t, u)$.

Theorem 1. ([9, Thm. 3]) *Let $\mathbb{K}$ be a field of characteristic 0 and consider two polynomials $f \in \mathbb{K}[u]$ and $Q \in \mathbb{K}[x, y_1, \dots, y_k, t, u]$, where $k \in \mathbb{N} \setminus \{0\}$. Let $a \in \mathbb{K}$ and $\Delta_a : \mathbb{K}[u][[t]] \rightarrow \mathbb{K}[u][[t]]$ be the divided difference operator $\Delta_a F(t, u) := (F(t, u) - F(t, a))/(u - a)$. Let us denote by Δ_a^ℓ the operator obtained by iterating ℓ times Δ_a . Then, there exists a unique solution $F \in \mathbb{K}[u][[t]]$ to the functional equation (2), and moreover $F(t, u)$ is algebraic over $\mathbb{K}(t, u)$.*

Applying Theorem 1 to our guiding example, there exists some polynomial $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, G(t)) = 0$. Namely, $G(t) = 1 + t + 6t^2 + 54t^3 + 594t^4 + \dots$ is annihilated by

$$R(t, z) = 81t^2 z^3 - 9t(9t - 2)z^2 + (27t^2 - 66t + 1)z - 3t^2 + 47t - 1.$$

Moreover G is the unique root of R in $\mathbb{Q}[[t]]$. Thus R provides a finite amount of data that encodes the sequence $(c_n)_{n \in \mathbb{N}}$. Now using R , we write the linear differential equation satisfied by $F(t, 1)$ and deduce the associated recurrence relation:

$$(81n^2 + 81n + 18) \cdot c_n - (4n^2 + 14n + 12) \cdot c_{n+1} = 0,$$

with $c_0 = 1, c_1 = 1$. Using this recurrence relation, computing $c_{3 \cdot 10^6}$ only takes 5 seconds in Maple. Solving the recurrence relation, one deduces $c_0 = 1$ and $c_n = \frac{4 \cdot 3^{n-1}}{(2n+2)(2n+1)} \binom{3n}{n}$, for all $n \geq 1$. This latter *closed formula* yields the asymptotic behavior

$$c_n \sim 3^{4n} 2^{-2n} (4\sqrt{3\pi} n^{-\frac{5}{2}} + O(n^{-\frac{7}{2}})), \text{ when } n \rightarrow \infty.$$

A central question is to solve equations like (2) that is, to compute annihilating polynomials for their solutions. We summarize the objective:

For a solution $F(t, u)$ of a DDE of the form (2), compute $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, F(t, a)) = 0$.

Previous algorithmic works. There exists a rich literature regarding the effective resolution of DDEs. The articles [9, 6, 7] already contain a complete state-of-the-art, we thus only give a quick overview of this literature. Let us start with DDEs such that the polynomial Q in (2) has degree 2 in x . When $k = 1$, Brown introduced in [13] what is now called the *quadratic method*. This method was generalized later by Bender and Canfield in [3] to a particular family of DDEs of arbitrary order, still with Q of degree 2 in x .

Also, the case where Q is linear in $x, y_1, \dots, y_k$ appears in many walks enumeration problems (e.g. [9, §3.1]). Solving DDEs in this case is usually done by applying the *kernel method*, introduced by Knuth in [20] and further studied by Banderier and Flajolet in [1]. Since the work of Bousquet-Mélou and Petkovšek [10], the linear case is considered as understood.

Regarding strategies based on the *guess-and-prove* paradigm (popularized in [22]), they have been notably studied by Zeilberger [26] in the case $k = 1$, and improved in this same case with Gessel in [19].

Bousquet-Mélou and Jehanne designed in [9] a general method which can be seen as a generalization of both the kernel and the quadratic methods. Their strategy consists in translating the resolution of a DDE into the resolution of a polynomial system admitting a solution with $F(t, a)$ as one of its coordinates. We refer to their algorithm as “duplication”.

Recently, intensive studies have been undertaken in [6] (resp. in [7]) in the direction of effectivity in order to design efficient algorithms for solving DDEs of order $k = 1$ (resp. of any order). On the one hand, Bostan, Chyzak, Safey El Din and the author designed two efficient algorithms based on effective algebraic geometry (we refer to these algorithms as “elimination” and “geometry”) and on the other hand they used these first geometry-driven algorithms in a new hybrid guess-and-prove approach (referred as “hybrid”).

*The algorithms are now ripe to be delivered turnkey to their natural recipients,
combinatorialists, via a Maple package and this tutorial paper.*

Contributions. The present paper contains three contributions. First, it describes the Maple package [DDE-Solver](#), dedicated to solving DDEs of any order that satisfy the (harmless) assumptions explicated in [7]. More explicitly, [DDE-Solver](#) contains the function

annihilating_polynomial.

This function takes as input a DDE of the form (2), its order k and a set of variables, and outputs a nonzero polynomial annihilating the series $F(t, a)$. Also, some options can be specified, namely: the *algorithm* that shall be used among “duplication”, “elimination”, “geometry” and “hybrid”; and an option *variable* explained in Section 5.

The second contribution is a careful practical comparison of these algorithms resulting in a table of timings.

The third contribution is the resolution with **annihilating_polynomial** of the DDE associated to the enumeration of 3-greedy Tamari intervals (resp. of 5-constellations) in 1 minute (resp. in 3 hours). It is the first time that these two equations are solved with an automated approach. We however underline that these DDEs are solved without using effective elimination theory in [12] for greedy Tamari intervals and in [11] for constellations.

To be clear, the aim of this tutorial paper is not to introduce new algorithms or to solve brand new DDEs but to introduce the Maple package **DDE-Solver** in a self contained tutorial paper which we believe is more understandable than the short paper [7] in which the algorithms were introduced. In contrast with [7], we added 4 new DDEs in the table of timings.

Structure of the paper. In Section 2, we recall why solving a DDE is reduced [9, Sec. 2] to eliminating variables in a system of polynomial equations. Also, we recall in Section 2 different modeling of the underlying geometric problem. These modelings are useful later in the paper. In Section 3, we state some preliminaries on effective algebraic geometry that are used in Section 4 when giving an overview on how the algorithms “duplication”, “elimination”, “geometry” and “hybrid” (resp. [7, §3, §5, §6, §4]) work. In Section 5, we explain how to get started with **DDE-Solver**, and how to use the options of **annihilating polynomial**. Finally in Section 6, we provide some timings that illustrate the efficiency of this new package on various DDEs from the literature [9, 4, 8, 12].

Notations. Denote by $\mathbb{K}$ a field of characteristic 0, by $\overline{\mathbb{K}}$ its algebraic closure, by $\mathbb{Q}$ the field of rational numbers and by $\mathbb{F}_p$ the finite field with p elements. For a fixed positive integer k , denote by $\underline{x}$ (resp. $\underline{u}$ and $\underline{z}$) the variables $x_1, \dots, x_k$ (resp. $u_1, \dots, u_k$ and $z_0, \dots, z_{k-1}$). Denote by $\mathbb{K}[x_1, \dots, x_k]$ the ring of polynomials in the variables $x_1, \dots, x_k$ with coefficients in $\mathbb{K}$. For $P \in \mathbb{K}[\underline{x}]$, denote by $V(P) \subset \overline{\mathbb{K}}^r$ its zero set. For any ideal $\mathcal{I} \subset \mathbb{K}[\underline{x}]$ and any set of polynomials $\mathcal{S} \subset \mathbb{K}[\underline{x}]$, denote by $V(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ (resp. by $V(\mathcal{S}) \subset \overline{\mathbb{K}}^k$) the zero set of $\mathcal{I}$ (resp. of $\mathcal{S}$). Still for $P \in \mathbb{K}[\underline{x}]$, denote by $\text{disc}_{x_j}(P)$ the discriminant [15, 16., §6, Ch.3] of P with respect to x_j and by $\text{LC}_{x_j}(P)$ the leading coefficient of P with respect to x_j . At last, denote by $\overline{\mathbb{K}}[[t^{\frac{1}{*}}]]$ the ring $\bigcup_{d \geq 1} \overline{\mathbb{K}}[[t^{\frac{1}{d}}]]$ of Puiseux series in t with positive fractional exponents.

2. FROM DDEs TO POLYNOMIAL SYSTEMS

Starting from a DDE of the form (2), we multiply it by the least power of $(u - a)$ such that we obtain a polynomial functional equation of the form

$$P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \quad (3)$$

for some polynomial $P \in \mathbb{Q}[x, z_0, \dots, z_{k-1}, t, u]$. Taking the derivative of (3) with respect to u yields by the chain rule

$$\begin{aligned} \partial_u F(t, u) \cdot \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) \\ + \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{aligned} \quad (4)$$

Consider the equation in u given by

$$\partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \quad (5)$$

Observe that the solutions of (5) that belong to $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ are also solutions, by using (4), of the equation in u given by

$$\partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0.$$

Thus any non constant solution in $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ of (5) is a solution in u of the system of constraints

$$\begin{cases} P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{cases} \quad u(u - a) \neq 0, \quad (6)$$

The below hypothesis needs to be satisfied in order to make the ongoing general strategy work. This assumption holds for the examples considered in this paper.

Global assumption: There exist k distinct solutions $U_1(t), \dots, U_k(t) \in \overline{\mathbb{Q}}[[t^{\frac{1}{k}}]]$ (GA) to the system of equations (6). Also, $U_1(t), \dots, U_k(t) \notin \overline{\mathbb{Q}}$.

In [7], the authors studied three geometric interpretations of (GA) that we recall below. The first one is introduced in [9], while the last two were introduced respectively in [7, §5, §6]. Also, it is a nontrivial consequence of the proof of [9, Thm. 3] that the series $\{\partial_u^{i-1} F(t, a), U_i(t), F(t, U_i(t))\}_{1 \leq i \leq k}$ considered in this paper are elements of $\overline{\mathbb{Q}(t)}$.

Duplication approach. This approach was introduced by Bousquet-Mélou and Jehanne in [9, Sec. 2] and works as follows. It results from (GA) that the following relations hold:

$$\forall 1 \leq i \leq k, \begin{cases} P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) &= 0, \\ \partial_x P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) &= 0, \\ \partial_u P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) &= 0, \end{cases}$$

and

$$\prod_{1 \leq i < j \leq k} (U_i(t) - U_j(t)) \cdot \prod_{1 \leq i \leq k} U_i(t) \cdot (U_i(t) - a) \neq 0.$$

In terms of polynomial equations, the above relations are equivalent to saying that the *duplicated* polynomial system $\mathcal{S}_{\text{dup}}$ defined by

$$\forall 1 \leq i \leq k, \begin{cases} P(x_i, z_0, \dots, z_{k-1}, t, u_i) &= 0, \\ \partial_x P(x_i, z_0, \dots, z_{k-1}, t, u_i) &= 0, \\ \partial_u P(x_i, z_0, \dots, z_{k-1}, t, u_i) &= 0, \end{cases} \quad m \cdot \prod_{1 \leq i < j \leq k} (u_i - u_j) \cdot \prod_{1 \leq i \leq k} u_i \cdot (u_i - a) - 1 = 0,$$

admits the nontrivial solutions

$$x_i = F(t, U_i(t)), u_i = U_i(t), z_{i-1} = \partial_u^{i-1} F(t, a), \text{ for } 1 \leq i \leq k. \quad (7)$$

Note that (7) uniquely determines the value associated to the variable m . Also, the polynomial system $\mathcal{S}_{\text{dup}}$ admits $3k + 1$ equations and unknowns (t is a parameter as we work over $\mathbb{Q}(t)$); that is, the polynomials in $\mathcal{S}_{\text{dup}}$ are seen as elements of the ring $\mathbb{Q}(t)[m, \underline{x}, \underline{z}, \underline{u}]$. Denote by $V(\mathcal{S}_{\text{dup}})$ the solution set of $\mathcal{S}_{\text{dup}}$ in $\overline{\mathbb{Q}(t)}^{k+3}$.

Elimination theory approach. This second approach was introduced in [7, Sec. 5]. The idea is that it follows from (6) and (GA) that the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, u]$ by:

$$\begin{cases} P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \\ \partial_x P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \\ \partial_u P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \end{cases} \quad u(u - a) \neq 0, \quad (8)$$

admits the k solutions $(x, u) = (F(t, U_i(t)), U_i(t)) \in \overline{\mathbb{Q}(t)}^2$, for $1 \leq i \leq k$. Note that these solutions have distinct u -coordinates. This observation can be reformulated geometrically in the following way. Consider the geometric projection $\pi : (x, u, \underline{z}) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (\underline{z}) \in \overline{\mathbb{Q}(t)}^k$, and denote by $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^{k+2}$ the solution set of the polynomial constraints (t is present but

still considered as a parameter)

$$\begin{cases} P(x, z_0, \dots, z_{k-1}, t, u) &= 0, \\ \partial_x P(x, z_0, \dots, z_{k-1}, t, u) &= 0, \quad u(u-a) \neq 0. \\ \partial_u P(x, z_0, \dots, z_{k-1}, t, u) &= 0, \end{cases} \quad (9)$$

For $\alpha \in \overline{\mathbb{Q}(t)}^k$, denote by $\#_u(\mathcal{X}, \alpha)$ the number of u -coordinates in $\pi^{-1}(\alpha) \cap \mathcal{X}$ that are not constants and define $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^k \mid \#_u(\mathcal{X}, \alpha) \geq k\}$. Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{F}_k(u, \mathcal{X})$.

Geometric approach. This last approach was introduced in [7, Sec. 6]. The idea is that in addition to having (6), it follows from (GA) that the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, z_1, u]$ (t is considered as a parameter) by:

$$\begin{cases} P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \\ \partial_x P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \quad u(u-a) \neq 0, \\ \partial_u P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) &= 0, \end{cases} \quad (10)$$

admits the k solutions $(x, z_1, u) = (F(t, U_i(t)), \partial_u F(t, a), U_i(t)) \in \overline{\mathbb{Q}(t)}^{k+1}$, for $1 \leq i \leq k$. Note that these solutions are distinct. Denote by $\tilde{z}_1$ the set of variables $z_0, z_2, \dots, z_{k-1}$. Consider the geometric projection $\pi_{\tilde{z}_1} : (x, u, \underline{z}) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (\tilde{z}_1) \in \overline{\mathbb{Q}(t)}^{k-1}$ and define

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{\tilde{z}_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{S}_k(\mathcal{X})$.

Summary. In Section 2, we defined for each of the three approaches a solution set which contains a point whose z_0 -coordinate is the series $F(t, a)$. The goal of Section 4 is to recall how the algorithms studied in [7, §3, §5, §6] compute a polynomial characterization of these sets and deduce from this characterization an annihilating polynomial of $F(t, a)$.

3. SOME PRELIMINARIES ON POLYNOMIAL TOOLS

The current section is devoted to the introduction of the tools used in Section 4.

3.1. ELIMINATION THEORY THROUGH GRÖBNER BASES

For $x_1, \dots, x_n$ some variables and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$, denote $\mathbf{x}^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. Also for $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, denote by $\langle \mathcal{S} \rangle$ the ideal generated by $\mathcal{S}$ in $\mathbb{K}[x_1, \dots, x_n]$.

Monomial orders and Gröbner bases. The idea of elimination monomial orderings is to attribute to a variable (or to a block of variables) which shall be eliminated a larger weight than the weights of the other variables. An example is the lexicographic order $\succ_{\text{lex}}$.

Definition 1. ([15, Def.3, §2, Ch.2]) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ be in $\mathbb{Z}_{\geq 0}^n$. We say that $\mathbf{x}^\alpha \succ_{\text{lex}} \mathbf{x}^\beta$ if the leftmost nonzero entry of the vector difference $\alpha - \beta \in \mathbb{Z}^n$ is positive.

Example 1. In $\mathbb{K}[x_1, x_2]$: $x_1^4 x_2^2 \succ_{\text{lex}} x_1^3 x_2^{10}$, $x_1^2 x_2 \succ_{\text{lex}} x_1$ and $x_1^2 \succ_{\text{lex}} x_2$.

Another useful family of monomial orders that we will refer to in Section 4 is the family of graded monomial orders. An example is the graded reverse lexicographic order $\succ_{\text{grevlex}}$.

Definition 2. ([15, Def.6, §2, Ch.2]) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ be in $\mathbb{Z}_{\geq 0}^n$. We say that $x^\alpha \succ_{\text{grevlex}} x^\beta$ if $|\alpha| = \sum_{i=1}^n \alpha_i > |\beta| = \sum_{i=1}^n \beta_i$, or if $|\alpha| = |\beta|$ and the rightmost nonzero entry of $\alpha - \beta \in \mathbb{Z}^n$ is negative.

Example 2. In $\mathbb{K}[x_1, x_2, x_3]$: $x_1^5 x_2^7 x_3 \succ_{\text{grevlex}} x_1^4 x_2^2 x_3^3$, $x_1 x_2^4 x_3^2 \succ_{\text{grevlex}} x_1^3 x_2 x_3^3$

We invite readers unfamiliar with monomial orders (see [15, Def.1, §2, Ch.2] for a general definition) to choose one of the two above monomial orders $\succ_{\text{lex}}$ or $\succ_{\text{grevlex}}$ for the next definition. Also, for $p \in \mathbb{K}[x_1, \dots, x_n]$ a polynomial and $\succ$ a monomial order on $\mathbb{K}[x_1, \dots, x_n]$, denote by $\text{LT}_\succ(p)$ the leading term of p with respect to the monomial order $\succ$.

Definition 3. ([15, Def.5, §5, Ch.2]) Fix a monomial order $\succ$ on the ring $\mathbb{K}[x_1, \dots, x_n]$. A finite subset $G = \{g_1, \dots, g_s\}$ of an ideal $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ different from $\{0\}$ is said to be a Gröbner basis of $\mathcal{I}$ for the order $\succ$ if $(\text{LT}(g_1), \dots, \text{LT}(g_n)) = \langle \text{LT}(g) \mid g \in \mathcal{I} \rangle$.

A fundamental property [15, Cor.6, §5, Ch.2] of Gröbner bases is, with the notations of Definition 3, that such a basis G always exists, and that G generates $\mathcal{I}$, that is $\langle G \rangle = \mathcal{I}$.

Gröbner bases and projections. The results below justify the use of Gröbner bases as a dedicated theoretical and computational tool, allowing one to characterize projections as solution sets of conjunctions of polynomial equations and inequations.

Theorem 2 (Elimination theorem). ([15, Thm.2, §1, Ch.3]) Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal. Denote by G a Gröbner basis of $\mathcal{I}$ with respect to the order $\succ_{\text{lex}}$. Then for every $0 \leq \ell \leq n-1$, the set $G_\ell = G \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$ is a Gröbner basis of the ideal $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$.

By Theorem 2, the set of polynomials G_ℓ finitely generates $\mathcal{I}_\ell$. Let us now introduce the geometric projection $\pi_\ell : (x_1, \dots, x_n) \in \overline{\mathbb{K}}^n \mapsto (x_{\ell+1}, \dots, x_n) \in \overline{\mathbb{K}}^{n-\ell}$. Recall that a Zariski closed set in $\overline{\mathbb{K}}^n$ is defined as the solution set of some polynomial equations defined in $\mathbb{K}[x_1, \dots, x_n]$.

Theorem 3 (Closure theorem). ([15, Thm.3, §2, Ch.3]) Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $V(\mathcal{I}) \subset \overline{\mathbb{K}}^n$ its zero set. Consider $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$. Then $V(\mathcal{I}_\ell)$ is the smallest Zariski closed set containing $\pi_\ell(V(\mathcal{I})) \subset \overline{\mathbb{K}}^{n-\ell}$.

Theorems 2 and 3 imply that characterizing the Zariski closure of the projection of a zero set $V(\mathcal{I})$ onto some coordinate subspace is done by computing a Gröbner basis for $\succ_{\text{lex}}$.

Theorem 4 (Extension theorem). ([15, Thm.2, §5, Ch.3]) Let $G = \{g_1, \dots, g_s\}$ be a Gröbner basis of $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ for the order $\succ_{\text{lex}}$. For each $1 \leq j \leq s$, consider

$$g_j = c_j(x_2, \dots, x_n) \cdot x_1^{N_j} + (\text{terms in which } x_1 \text{ has degree} < N_j),$$

where $N_j \geq 0$ and $c_j \in \mathbb{K}[x_2, \dots, x_n]$ is nonzero. Assume $\alpha = (\alpha_2, \dots, \alpha_n) \in V(\mathcal{I} \cap \mathbb{K}[x_2, \dots, x_n])$ is a partial solution with the property that $\alpha \notin V(c_1, \dots, c_s)$. Then

$$\{f(x_1, \alpha) \mid f \in \mathcal{I}\} = \{g_{j_0}(x_1, \alpha)\},$$

where $g_{j_0} \in G$ satisfies $c_{j_0}(\alpha) \neq 0$ and g_{j_0} has minimal degree in x_1 among all elements g_j with $c_j(\alpha) \neq 0$. Furthermore, if $g_{j_0}(\alpha_1, \alpha) = 0$ for $\alpha_1 \in \overline{\mathbb{K}}$, then $(\alpha_1, \alpha) \in V(\mathcal{I})$.

A consequence of Theorem 4 is that characterizing projections (and not their Zariski closure) can be done by considering the disjunction of inequations associated to the non simultaneous vanishing of the $c_1, \dots, c_s$ of Theorem 4: for each condition in this disjunction, one shall add the vanishing of all the elements of G_ℓ of Theorem 2.

3.2. COUNTING SPECIFIC SOLUTIONS

Let n, s be positive integers. Consider the notation $\mathbf{x} = x_1, \dots, x_n$ and $\mathbf{y} = y_1, \dots, y_s$. Let $\mathcal{I} \subset \mathbb{K}[\mathbf{x}]$ be a radical ideal such that its zero set $V(\mathcal{I}) \subset \overline{\mathbb{K}}^n$ is finite. For $\alpha \in V(\mathcal{I})$, denote by $[x_1](\alpha)$ the x_1 -coordinate of α . Denote the projection $\pi_{x_1} : (\mathbf{x}) \in \overline{\mathbb{K}}^n \mapsto x_1 \in \overline{\mathbb{K}}$. Let $g \in \mathbb{K}[\mathbf{y}][z]$ and denote $\text{LC}_z(g) \in \mathbb{K}[\mathbf{y}]$ the leading coefficients of g in z .

This current subsection introduces tools for answering the two problems below.

Problem 5. Let ℓ be a positive integer. Characterize with polynomial equations defined in $\mathbb{K}[\mathbf{x}]$ the set $\{\beta \in V(\mathcal{I}) \mid \pi_{x_1}^{-1}([x_1](\beta)) \cap V(\mathcal{I}) \geq \ell\}$.

Problem 6. Let $1 \leq \ell \leq \deg_z(g)$. Characterize with polynomial inequations defined in $\mathbb{K}[\mathbf{y}]$ the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_z(g))$ such that $g(\mathbf{y} = \beta, z)$ has at least ℓ distinct roots.

Answer to Problem 5. Denote $\mathcal{A} := \mathbb{K}[\mathbf{x}]/\mathcal{I}$. It results from the finiteness of $V(\mathcal{I})$ that $\mathcal{A}$ has finite dimension as a $\mathbb{K}$ -vector space [15, Thm.6, §3, Ch.5]. Define $m_{x_1} : f \in \mathcal{A} \mapsto x_1 \cdot f \in \mathcal{A}$ to be the multiplication map by x_1 in $\mathcal{A}$, and denote by $\chi_{x_1} \in \mathbb{K}[T]$ its characteristic polynomial. The following results from the radicality of $\mathcal{I}$ and from [16, Prop.2.7, §2, Ch.4].

Fact 1. We have the equality $\chi_{x_1} = \prod_{\alpha \in V(\mathcal{I})} (T - [x_1](\alpha))$.

As a consequence of Fact 1, we obtain the following set equality:

$$\begin{aligned} \{\beta \in V(\mathcal{I}) \mid \#\pi_{x_1}^{-1}([x_1](\beta)) \cap V(\mathcal{I}) \geq \ell\} \\ = \{\beta \in \overline{\mathbb{K}}^n \mid \beta \in V(\mathcal{I}) \wedge \chi_{x_1}([x_1](\beta)) = 0 \wedge \dots \wedge \partial_T^{\ell-1} \chi_{x_1}([x_1](\beta)) = 0\}. \end{aligned} \quad (11)$$

Equality (11) provides a conjunction of polynomial equations that answers Problem 5.

Answer to Problem 6. Define the Hermite quadratic form associated with g by

$$H_g : (f, h) \in (\mathbb{K}(\mathbf{y})[z]/\langle g \rangle)^2 \mapsto \text{Trace}(m_{f \cdot h}) \in \mathbb{K}(\mathbf{y}),$$

where $\text{Trace}(\cdot)$ is the trace operator and $m_{f \cdot h}$ is the multiplication map by $f \cdot h$ in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$. Also, denote by $M_{H_g} \in \mathbb{K}(\mathbf{y})^{\deg_z(g) \times \deg_z(g)}$ the matrix of H_g in the basis $\{1, z, \dots, z^{\deg_z(g)-1}\}$.

The following fact is an immediate consequence of [16, Thm.5.2, §5, Ch.2] and of the observation that, by performing euclidian divisions by g in $\mathbb{K}(\mathbf{y})[z]$, the denominator of the image of z^i in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$ can only be a power of $\text{LC}_z(g)$.

Fact 2. The denominators in the matrix M_{H_g} are powers of $\text{LC}_z(g)$. Moreover if $1 \leq \ell \leq \deg_z(g)$, then the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_z(g))$ at which $g(\mathbf{y} = \beta, z)$ admits less than ℓ distinct solutions are the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_z(g))$ at which the $\ell \times \ell$ -minors of M_{H_g} all vanish.

Fact 2 implies that the conjunction of polynomial inequations answering Problem 6 is given by the non vanishing of $\text{LC}_z(g)$ and by the non vanishing of the $\ell \times \ell$ -minors of M_{H_g} .

3.3. CHANGE OF MONOMIAL ORDERING

For n generic elements of $\mathbb{K}[x_1, \dots, x_n]$ of degree d , computing a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$ has an arithmetic cost which is in $d^{O(n)}$ [2, Prop.1].

Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal whose solution set is finite. The following fact mentions an algorithm called FGLM [17] that on input a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$ outputs a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{lex}}$. The complexity of this algorithm is, under

some genericity assumptions, in $\tilde{O}(D^\omega)^1$ (see [21]), where D is the cardinality of $V(\mathcal{I})$ ($\mathcal{I}$ is assumed radical). By the Bézout bound, and with the notations of the above paragraph, we have the bound $D \leq d^n$, this makes the following fact useful for computing in $d^{O(n)}$ a Gröbner bases of $\mathcal{I}$ for a $\succ_{\text{lex}}$ order.

Fact 3. *On input an ideal $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ whose solution set $V(\mathcal{I}) \subset \overline{\mathbb{K}}^n$ is finite, there exists an algorithm that computes a Gröbner basis $G_{\succ_{\text{grevlex}}}$ of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$ and that turns it into a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}$ for the order $\succ_{\text{lex}}$.*

With the notations as above and without using the algorithm FGLM [17] mentioned in Fact 3, the computation of a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{lex}}$ has an arithmetic cost bounded by $C_2 d^{C_3 n^3}$ [14], for $C_2, C_3 \in \mathbb{Z}_{\geq 0}$.

4. FUNCTION IMPLEMENTED IN DDE-SOLVER

The function implemented in **DDE-Solver** is:

annihilating_polynomial

We explain in this section the algorithms on which this function, and its options, rely.

The function **annihilating_polynomial** computes an annihilating polynomial of the series $F(t, a)$. Four algorithms can be applied: “duplication”, “elimination”, “geometry”, “hybrid”. These algorithms are studied respectively in [7, §3, §5, §6, §4]. As the algorithm “hybrid” relies on the three other algorithms, we first describe “duplication”, “elimination” and “geometry”.

In Section 2, we defined solution sets that contain a point whose z_0 -coordinate is $F(t, a)$:

- “duplication”: $V(\mathcal{S}_{\text{dup}})$ contains the point defined by (7),
- “elimination”: $\mathcal{F}_k(u, \mathcal{X})$ contains the point $(F(t, a), \dots, \partial_u^{k-1} F(t, a))$,
- “geometric”: $\mathcal{S}_k(\mathcal{X})$ contains the point $(F(t, a), \dots, \partial_u^{k-1} F(t, a))$.

We introduce a finiteness assumption that holds for the examples studied in this paper.

Finiteness assumption: The sets $V(\mathcal{S}_{\text{dup}})$, $\mathcal{F}_k(u, \mathcal{X})$ and $\mathcal{S}_k(\mathcal{X})$ are finite. **(FA)**

General strategy. Under **(FA)**, the general spirit of the algorithms from [7, §3, §5, §6] is:

- To compute a disjunction of polynomial equations whose solution set is the set of interest,
- To eliminate all variables except (z_0, t) from the polynomial characterization (i).

Let us apply the above steps to the three approaches from Section 2.

4.1. OPTION “DUPLICATION”

Step (i) of the general strategy: The polynomial system $\mathcal{S}_{\text{dup}}$ characterizes $V(\mathcal{S}_{\text{dup}})$.

Step (ii) of the general strategy: Denote by $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, \underline{x}, \underline{z}, \underline{u}]$ the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$. By assumption **(FA)**, it is possible to apply to $\mathcal{I}_{\text{dup}}$ the algorithm from Fact 3. This algorithm outputs, for a proper choice of variable ordering, a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}_{\text{dup}}$ which contains an element of $\mathbb{Q}(t)[z_0]$. Applying Theorem 3 to the projection onto the z_0 -coordinate space together with Theorem 2 yields, under **(FA)**, that this polynomial is

¹We denote by ω the constant of multiplication matrix that is, multiplying two matrices in $\mathbb{K}^{n \times n}$ can be done by using $\tilde{O}(n^\omega)$ arithmetic operations in $\mathbb{K}$, where $\tilde{O}(\cdot)$ is a $O(\cdot)$ that hides polylogarithmic factors.

nonzero. Denote by $R \in \mathbb{Q}[t, z_0]$ its numerator. By [7, Prop. 2], we have $R(t, F(t, a)) = 0$.

We refer the reader to Annex 7.1 for the resolution of 3-constellations via this method.

4.2. OPTION “ELIMINATION”

Recall from Section 2 that $\mathcal{F}_k(u, \mathcal{X}) = \{\alpha \in \overline{\mathbb{Q}(t)}^k \mid \#_u(\mathcal{X}, \alpha) \geq k\}$. The algorithm from [7, §5] requires the additional technical assumption that for any $\alpha \in \overline{\mathbb{Q}(t)}^k$, we have $\#_u(\mathcal{X}, \alpha) < +\infty$. We do not make it explicit for sake of simplicity. Under this technical assumption and (FA), the algorithm designed in [7, §5] works as follows.

Step (i) of the general strategy:

- (1) Compute using Theorems 2 and 4 successive disjunctions of conjunctions of polynomial equations and inequations defined in $\mathbb{K}(t)[u, \underline{z}]$ whose solution sets are successively the projections onto the $(u, \underline{z})$ -coordinate space and onto the $(\underline{z})$ -coordinate space.

The union of the polynomial constraints in these disjunctions have the form $\{I, E\}$, where I is a set of inequations defined in $\mathbb{Q}(t)[u, \underline{z}]$ and E is a set of equations defined in $\mathbb{Q}(t)[u, \underline{z}]$. Without loss of generality, we assume that the polynomials in E are a Gröbner basis of the ideal they generate, for the order $\succ_{\text{lex}}$ with u greater than $\underline{z}$.

- (2) Apply Theorem 4 to E : at $\underline{z} = \alpha \in \overline{\mathbb{Q}(t)}^k$ fixed, the cardinality condition in the definition of $\mathcal{F}_k(u, \mathcal{X})$ is equivalent to studying for a polynomial g_{j_0} in E :
 - if g_{j_0} has its leading coefficient in u that does not vanish at $\underline{z} = \alpha$ and, by Theorem 4, if g_{j_0} is the polynomial in E of minimal degree in u that satisfies this property: checking this minimality condition yields a disjunction of polynomial equations and inequations defined in $\mathbb{Q}(t)[\underline{z}]$,
 - for this g_{j_0} , it remains to add a conjunction of polynomial conditions in $\mathbb{Q}(t)[u, \underline{z}]$ so that $g_{j_0}(u, \underline{z} = \alpha)$ has at least k distinct solutions:
If $\deg_u(g_{j_0}) < k$, adds the vanishing of all the coefficients (in u) of g_{j_0} to E ; else adds the conjunction of inequations given by the non vanishing of the $k \times k$ -minors of $M_{H_{g_{j_0}}}$ from Fact 2.
- (3) Steps 1 and 2 above computed a disjunction of polynomial equations and inequations defined in $\mathbb{Q}(t)[u, \underline{z}]$ whose solution set in $\overline{\mathbb{Q}(t)}^{k+1}$ is $\mathcal{F}_k(u, \mathcal{X})$. As by (FA) this set is finite, this polynomial characterization can be turned into a disjunction of conjunction of polynomial equations by using Rabinowitsch trick² in order to remove the solution set given by the inequations: the introduced variables are eliminated using Theorem 2. We denote by $\mathcal{D}_{\text{elim}}$ the so-obtained disjunction.

Step (ii) of the general strategy: We perform this step in the same spirit as what we did in Section 4.1. By (FA), the solution set of $\mathcal{D}_{\text{elim}}$ is finite. We can thus apply the algorithm from Fact 3 to each disjunction in $\mathcal{D}_{\text{elim}}$. Each application of this algorithm allows to compute a nonzero polynomial of $\mathbb{Q}(t)[z_0]$: this polynomial might be equal to 1 when there is no solution to the studied conjunction. We consider the numerators of all these polynomials and denote by $R \in \mathbb{Q}[t, z_0]$ their product. By [7, Prop. 5.5], we have $R(t, F(t, a)) = 0$.

²Instead of considering an inequation $Q \neq 0$ (for a given polynomial $Q \in \mathbb{K}[x_1, \dots, x_n]$), introduce an extra variable m and consider the equation $m \cdot Q - 1 = 0$ defined in $\mathbb{K}[m, x_1, \dots, x_n]$.

We refer the reader to Annex 7.2 for the resolution of 3-constellations via this method.

4.3. OPTION “GEOMETRY”

Recall from Section 2 that

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{z_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

The algorithm from [7, §6] requires many additional technical assumptions (see the beginning of [7, §6]). We do not make them explicit for sake of simplicity. In the rest of this section, denote by $\succ_{\text{bgrevlex}}$ the block monomial order over $\mathbb{Q}(t)[m, x, u, z]$ defined as follows:

- We use the monomial order $\succ_{\text{grevlex}}$ on each of the two blocks $\{m, x, u, z_1\}$ and $\{z_1\}$,
- Two monomials are compared w.r.t. the variables m, x, u, z_1 . In case of equality, they are compared w.r.t. the variables z_1 .

Example 3. For $k = 3$:

$$x^3 u z_1^2 m \succ_{\text{bgrevlex}} z_0^2 z_2^3, \quad x^3 u z_1^2 m z_2 \succ_{\text{bgrevlex}} x^3 u z_1^2 m z_0 \quad \text{and} \quad z_0^4 \succ_{\text{bgrevlex}} z_0^2 z_2.$$

Under these assumptions and (FA), the algorithm designed in [7, §6] works as follows.

Step (i) of the general strategy:

- (1) Compute a Gröbner basis G of the ideal $\mathcal{I} := \langle P, \partial_x P, \partial_u P, m \cdot u(u - a) - 1 \rangle \subset \mathbb{Q}(t)[m, x, u, z]$ for the order $\succ_{\text{bgrevlex}}$.
- (2) Compute, using normal form computations modulo $\mathcal{I}^3$, the matrix M_{z_1} of the multiplication map m_{z_1} in the quotient ring $\mathbb{Q}(t, z_1)[m, x, u, z_1]/j(\mathcal{I})$, where $j(\mathcal{I})$ is the image of the injective map $j : \mathbb{Q}(t)[m, x, u, z] \rightarrow \mathbb{Q}(t, z_1)[m, x, u, z_1]$.
- (3) Compute the characteristic polynomial $\chi_{z_1} \in \mathbb{Q}(t, z_1)[T]$. Using the assumptions of [7, §6], we can replace χ_{z_1} by its numerator with respect to the variables z_1 , so that we now have $\chi_{z_1} \in \mathbb{Q}(t)[z_1, T]$.
- (4) Denote by $\mathcal{D}_{\text{geom}}$ the set of equations given by the vanishing of the polynomials in G and of the new conditions $(\chi_{z_1})|_{T=z_1} = 0 \wedge \dots \wedge (\partial_T^{k-1} \chi_{z_1})|_{T=z_1} = 0$.

Step (ii) of the general strategy: We perform this step in the same spirit as we did in Section 4.1. By (FA), the solution set of $\mathcal{D}_{\text{geom}}$ is finite. We thus apply the algorithm from Fact 3 to the ideal generated by the polynomials associated with the equations in $\mathcal{D}_{\text{geom}}$. This algorithm outputs a nonzero polynomial of $\mathbb{Q}(t)[z_0]$. Consider the numerator of this polynomial and denote it by $R \in \mathbb{Q}[t, z_0]$. By [7, Prop. 6.4], we have $R(t, F(t, a)) = 0$.

We refer the reader to Annex 7.3 for the resolution of 3-constellations via this method.

4.4. OPTION “HYBRID”

The *hybrid guess-and-prove* algorithm [7, §4] works as follows:

- (1) Compute bounds b_{z_0}, b_t such that $\deg_t(R) \leq b_t$ and $\deg_{z_0}(R) \leq b_{z_0}$, for some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, a)$ (we detail this step in the next section),
- (2) Compute the truncated series $F(t, a) \bmod t^{2b_t b_{z_0} + 1}$,
- (3) Guesse a polynomial $M \in \mathbb{Q}[t, z_0]$ such that $M(t, F(t, a)) = O(t^{(b_t+1)(b_{z_0}+1)-1})$,
- (4) Check that $M(t, F(t, a)) = O(t^{2b_t b_{z_0} + 1})$.

When step 4 is satisfied, applying [7, Prop.5] yields $M(t, F(t, a)) = 0$.

We refer the reader to Annex 7.4 for the resolution of 3-constellations via this method.

³Normal form computations modulo ideals are generalizations of euclidean divisions by polynomials.

4.5. DETAILS OF IMPLEMENTATIONS

As we target practical efficiency, the implementations of the options “duplication”, “elimination” and “geometry” incorporate the following key improvements from computer algebra. Recall that the output of these methods is a nonzero $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, a)) = 0$.

To reduce the computations of $\mathbb{Q}(t)$ to $\mathbb{Q}$, we perform evaluation-interpolation on t .

Also, we rely on fast multi-modular arithmetic. This consists in applying multiple times a given algorithm with the base field $\mathbb{Q}$ replaced by successive distinct prime fields $\mathbb{F}_p$. These computations output various images mod p of the same $R \in \mathbb{Q}[t, z_0]$. From there, one lifts the modular coefficients over $\mathbb{Q}$ by applying the Chinese Remainder Theorem (CRT) together with rational numbers reconstruction [18, §5.10].

Implementation of the option “hybrid”. Computing the bounds (b_t, b_{z_0}) in step 1 is done using the option “elimination” of Section 4.2. More precisely, computing b_{z_0} is done by specializing t at a random value θ of some random prime field $\mathbb{F}_p$ and performing the option “elimination” with base field $\mathbb{F}_p$: it outputs $R(\theta, z_0)$ and yields $b_{z_0} = \deg_{z_0}(R(\theta, z_0))$ (a similar computation is done for computing b_t). The generation of terms is done using a divide and conquer approach which computes the first terms of $F(t, u) \bmod t^{2b_t b_{z_0} + 1}$ before specializing them to $u = a$. Finally, the guessing is done by computing Hermite-Padé approximants. For this, we use the function `seriestoalgeq` of the Maple package `gfun` [24].

5. GETTING STARTED

5.1. INSTALLATION AND CUSTOMIZATION

Files provided: `DDE-Solver` is available in the “.mla” format. If one wishes to customize the package, the Maple scripts “ddesolver.mpl” and “build.mpl” can be downloaded on the dedicated [github webpage](#). Used as follows, they allow to modify the package by modifying “ddesolver.mpl” and to generate the corresponding version of “ddesolver.mla” by executing “build.mpl”.

Customization of `DDE-Solver`: Replacing “PATH/TO” in the Maple file “build.mpl” by the relevant path to the file where “ddesolver.mpl” and “build.mpl” are located allows one, by executing “build.mpl”, to generate a new version of “ddesolver.mla”.

Loading `DDE-Solver`: The Maple variable `libname` shall be set so that “ddesolver.mla” is located in a visible place.

```
libname := "/home/notarantonio/ddesolver/lib", libname:
```

Once `libname` has been correctly set up, one executes in Maple

```
with(ddesolver);
```

in order to load and use the package.

5.2. USING `DDE-SOLVER`

Input/Output syntax. The arguments of `annihilating_polynomial` are (P, k, var) , where $P \in \mathbb{Q}[x, z_0, \dots, z_{k-1}, t, u]$ is the polynomial in (3), k is the order of the considered DDE and `var`

is $[x, z_0, \dots, z_{k-1}, t, u]$ (precisely in that order).

The output of **annihilating_polynomial**(P, k, var) is some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, a)) = 0$.

Example 4. Consider (1). Multiplying (1) by $(u - 1)^2$ yields

$$0 = (u - 1)^2(1 - F(t, u) + tuF(t, u)^3) + tu(u - 1)(2F(t, u) + F(t, 1))(F(t, u) - F(t, 1)) \\ + tu(F(t, u) - F(t, 1) - (u - 1)\partial_u F(t, 1)).$$

Thus $P := (u - 1)^2(1 - x + tux^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1)$ and $k := 2$. Continuing the analysis with Maple

```
P := (u-1)^2(1-x+tux^3)+tu(u-1)(2x+z0)(x-z0)+tu(x-z0-(u-1)z1);
with(ddesolver):
annihilating\_polynomial(P, 2, [x, z0, z1, t, u]);
```

outputs

$$(16tz_0^2 - 8tz_0 + t - 16) \\ (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

Thus $R := (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$ is an annihilating polynomial of $F(t, 1)$. A direct analysis on the solutions of $R(t, z_0)$ that are finite at $t = 0$ shows that the second factor of R is the minimal polynomial of $F(t, 1)$.

Options. It is possible to benefit in practice from two options: the choice of the *algorithm*, and the choice of the *variable* (either t or z_0) on which we perform evaluation–interpolation.

Choice of the algorithm. Four algorithms are implemented: “duplication”, “elimination”, “geometry” (only when $k = 2$) and “hybrid”. By default, the algorithm used is “elimination”.

Choice of the variable. The two choices are t and z_0 . The default choice is t .

The choice of the algorithm and of the variable on which we perform the evaluations is made by executing

annihilating_polynomial($P, k, \text{var}, \text{algorithm}, \text{variable}$).

Note that these two options must be either not specified at all, or specified in the same call.

Example 5. Continuing Example 4 with *algorithm* = “geometry” and *variable* = z_0 yields

```
annihilating\_polynomial(P, 2, [x, z0, z1, t, u], ‘‘geometry’’, z0);
```

$$(16tz_0^2 - 8tz_0 + t - 16) \\ (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

We refer to the next section for a discussion on the choice of the parameter variable.

6. EXAMPLES

6.1. IMPACT OF THE OPTION variable

The motivation of this study is the following. Assume that we perform evaluation–interpolation on z_0 . The chosen algorithm (e.g. “elimination” from Section 4.1) will be run a first time with t specialized (in order to get the degree in z_0 of the output $R \in \mathbb{Q}[t, z_0]$), then

it will be run multiple times at z_0 specialized (in order to evaluate–interpolate R w.r.t z_0). Hence if the algorithm is faster at t specialized than at z_0 specialized (and if the partial degrees of the output R are “close”), then the choice of the evaluation–interpolation variable has a significant impact on the timings. This is what happens below.

Example 6. Consider the DDE of the enumeration of 3-Tamari lattices [8, Prop. 8]:

$$F(t, u) = u + tuF(t, u) \frac{F(t, u) \frac{F(t, u) - F(t, 1)}{u-1} - F(t, 1) \partial_u F(t, 1)}{u-1} - \frac{F(t, 1)^2 \partial_u^2 F(t, 1)}{2} - F(t, 1) \partial_u F(t, 1)^2, \quad (12)$$

Multiplying (12) by $(u-1)^3$ yields the functional equation $P(F(t, u), F(t, 1), \partial_u F(t, 1), t, u) = 0$, where $P := -tu^3xz_0^2z_2 - 2tu^3xz_0z_1^2 - 2tu^2x^2z_0z_1 + 2tu^2xz_0^2z_2 + 4tu^2xz_0z_1^2 + 2tux^4 - 2tux^3z_0 + 2tux^2z_0z_1 - tuxz_0^2z_2 - 2tuxz_0z_1^2 + 2u^4 - 2u^3x - 6u^3 + 6u^2x + 6u^2 - 6ux - 2u + 2x \in \mathbb{Q}[x, z_0, z_1, z_2, t, u]$. We thus have in Maple

```
P := -tu^3xz0^2z2-2tu^3xz0z1^2-2tu^2x^2z0z1+2tu^2xz0^2z2+4tu^2
      xz0z1^2+2tux^4-2tux^3z0+2tux^2z0z1-tuxz0^2z2-2tuxz0z1^2+
      2u^4-2u^3x-6u^3+6u^2x+6u^2-6ux-2u+2x;
with(ddesolver):
time(annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u]));
      # variable = t (implicitly chosen)

2089.599

time(annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u],
      ‘elimination’, z0)); # variable = z0

141.897

R := annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u],
      ‘elimination’, z0);
R := t^5z0^16+135t^4z0^13+1024t^4z0^12+7290t^3z0^10-1762560t^3z0^9
      +393216t^3z0^8+196830t^2z0^7+111694464t^2z0^6+580976640t^2z0^5
      +t(67108864t+2657205)z0^4-661978656tz0^3+4721836032tz0^2
      +(-8371830784t+14348907)z0+4294967296t-14348907
```

6.2. PRACTICAL RESULTS OF THE FUNCTION ANNIHILATING_POLYNOMIAL

We gather in Table 6.2 some timings obtained after using the function

annihilating_polynomial.

Each column is associated with a DDE taken in the literature [4, 9, 8, 12]. For each of these DDEs, we precise: its order k , the variable on which we perform evaluation–interpolation, the algorithm which is used in the call to **annihilating_polynomial**, and finally the bi-degree of the output polynomial $R \in \mathbb{Q}[t, z_0]$.

From Table 1, we strengthen the message of Section 6.1 and draw some conclusions:

- *Choice of the algorithm* The best algorithm to choose depends in an important way on the studied DDE. Even among DDEs of the same family (e.g. DDEs [4], [5] and [8]) the answer is not clear: the first two are solved faster with the option

algorithm = “elimination” while the last one is solved faster with algorithm = “hybrid”. The efficiency of this last option algorithm = “hybrid” comes from the fact that the output polynomial has small bidegree (3, 9), thus allowing a fast computation of the first terms of $F(t, 1)$; and at the same time, that the computations are hard with the other algorithms.

This impossibility to predict which algorithm is more efficient for an input DDE justifies that we let the user choose the algorithm that shall be applied.

- *Choice of the variable* As in Section 6.1, the table shows that the choice of the variable (z_0 or t) on which we perform evaluation–interpolation is important. Let us pick one example to illustrate it. For the DDE [3], performing evaluation–interpolation over t with algorithm = “geometry” takes 2 minutes while it takes 54 minutes when performing evaluation–interpolation over z_0 . With the notations of Section 4.3, the reason is that at z_0 specialized to some random $\theta \in \mathbb{F}_p$, the main computational cost (in addition to the Gröbner basis computation) is to compute the discriminant w.r.t. T of $(\chi_{z_1})|_{z_0=\theta} \in \mathbb{Q}[t, T]$, whose partial degrees are 6 in T and 423 in t . In the same time at t specialized to some random $\nu \in \mathbb{F}_p$, the polynomial $(\chi_{z_1})|_{t=\nu} \in \mathbb{Q}[z_0, T]$ has degree 15 in z_0 and 6 in T : it yields a faster computation of its discriminant w.r.t. T .
As all these intermediate computational data are hard to predict from the input DDE, we let the user choose on which variable we shall perform evaluation–interpolation.
- *Solving DDEs previously out of reach!* The DDEs [7], [8] are solved via theoretical arguments in (resp.) [12, 11]. Here, it is the first time that they are solved in an automatized way!

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Data	[1]		[2]		[3]		[4]		[5]		[6]		[7]		[8]	
k	2		2		2		2		3		3		3		4	
eval–interp variable	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t
“duplication”	0.5s	0.7s	1.6s	2.3s	41m*	10m*	2.7s	1.5s	2m45s	2m50s	13h	27h	∞	∞	∞	∞
“elimination”	0.2s	0.3s	0.4s	0.4s	1h20m*	4m*	1.7s	1.2s	1m10s	45s	2m20s	35m	1m	7m30s	2d19h	2d
“geometry”	2s	2.2s	1.6s	0.9s	54m*	2m*	0.8s	0.7s	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
“hybrid”	50s		2m		∞		1.4s		18s		1h42m		34s		2h41m	
$(\deg_t(R), \deg_{z_0}(R))$	(4, 9)		(2, 3)		(132, 6)		(3, 5)		(3, 7)		(5, 16)		(2, 4)		(3, 9)	

FIGURE 2. Practical results of **annihilating_polynomial** on DDEs from the literature.

Computations are done in Maple on a computer equipped with Intel® Xeon® Gold CPU 6246R v4 @ 3.40GHz and 1.5TB of RAM with 1 thread. Gröbner bases computations are performed using the C library [msolve](#) [5], implemented by Berthomieu, Eder and Safey El Din.

- [1]: Enumeration of 2-Tamari lattices, [8, Prop. 8] for $m = 3$,
 - [2]: Enumeration of 2-greedy Tamari intervals, [12, Prop. 3.1] for $m = 2$,
 - [3]: Enumeration of non-separable near-triangulations in which all intern vertices have degree at least 5, [4, Prop. 4.3, Eq. 22],
 - [4]: Enumeration of 3-constellations, [9, Equation 29],
 - [5]: Enumeration of 4-constellations, [9, Prop. 12] for $m = 4$,
 - [6]: Enumeration of 3-Tamari lattices, [8, Prop. 8] for $m = 4$,
 - [7]: Enumeration of 3-greedy Tamari intervals, [12, Prop. 3.1] for $m = 3$,
 - [8]: Enumeration of 5-constellations. [9, Prop. 12] for $m = 5$.
-
- ∞ : The computations did not finish within 5 days,
 - $\times$: The algorithm is not implemented for $k > 2$,
 - $*$: We added the condition $m' \cdot (1 + t^3)(1 - t^3)t - 1 = 0$ to the input polynomial constraints.

7. APPENDIX

7.1. SOLVING 3-CONSTELLATIONS USING SECTION 4.1

The DDE [9, Eq. 29] associated with the enumeration of 3-constellations is given by

$$F(t, u) = 1 + tuF(t, u)^3 + tu(2F(t, u) + F(t, 1)) \frac{F(t, u) - F(t, 1)}{u - 1} + tu \frac{F(t, u) - F(t, 1) - (u - 1)\partial_u F(t, 1)}{(u - 1)^2} \quad (13)$$

Multiplying (13) by $(u - 1)^2$ yields $P(F(t, u), F(t, 1), \partial_u F(t, 1), t, u) = 0$, where

$$P := (u - 1)^2(1 - x + tx^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1) \in \mathbb{Q}[x, z_0, z_1, t, u].$$

Define the polynomial system

$$\begin{aligned} \mathcal{S}_{\text{dup}} := & (P(x_1, z_0, z_1, t, u_1) = 0, \partial_x P(x_1, z_0, z_1, t, u_1) = 0, \partial_u P(x_1, z_0, z_1, t, u_1) = 0, \\ & P(x_2, z_0, z_1, t, u_2) = 0, \partial_x P(x_2, z_0, z_1, t, u_2) = 0, \partial_u P(x_2, z_0, z_1, t, u_2) = 0, \\ & m \cdot (u_1 - u_2) \cdot (u_1 - 1) \cdot (u_2 - 1) \cdot u_1 \cdot u_2 - 1 = 0), \end{aligned}$$

and set $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, x_1, x_2, z_0, z_1, u_1, u_2]$ to be the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$.

The algorithm described in Section 4.1:

- (1) Computes a generator $R \in \mathcal{I}_{\text{dup}} \cap \mathbb{Q}(t)[z_0]$ by applying Fact 3 and finds

$$R = (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)(16tz_0^2 - 8tz_0 + t - 16).$$

7.2. SOLVING 3-CONSTELLATIONS USING SECTION 4.2

Define $\mathcal{S} := (P, \partial_x P, \partial_u P, m \cdot u(u - 1) - 1)$, for P as in 7.1. Denote by $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^5$ the solution set of the conditions $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u - 1) \neq 0$ in $\overline{\mathbb{Q}(t)}^4$.

Goal: Characterize $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^2 \mid \#_u(\mathcal{X}, \alpha) \geq 2\}$ as the solution set of some polynomial constraints and deduce a nonzero $R \in \mathbb{Q}[t, z_0]$ such that

$$R(t, F(t, 1)) = 0.$$

The algorithm described in Section 4.2:

- (1) Characterizes the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space:

```
G := Groebner[Basis](S, lexdeg([m, x], [u, z0, z1, t])):
E := Groebner[Basis](remove(has, G, {m, x}), lexdeg([u], [z0, z1, t])):
Leading := select(has, G, {m, x}):
```

The first element of Leading has the form

$$1534(u - 1) \cdot x + (\text{polynomial in } u, z_0, z_1, t).$$

As $\mathcal{X} \cap \{u = 1\} = \emptyset$, it results from the extension theorem that the zero set of E in $\overline{\mathbb{Q}(t)}^3$ is precisely the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space.

- (2) Identifies algebraic conditions in E describing the cardinality of the fiber in $\mathcal{F}_k(u, \mathcal{X})$:

```

# Dealing with the polynomials of degree < 2 in u in E:
NewConditions := [];
for i to nops(E) do
  if degree(E[i], u) < 2 then
    NewConditions := [op(NewConditions), coeffs(E[i], u)]:
  fi:
od:
E := Groebner[Basis]([op(E), op(NewConditions)],
  lexdeg([u], [z0, z1, t])):
# Dealing with the polynomials of degree > 1 in u in E:
map(p -> degree(p, u), E);
# prints the degrees in u of the polynomials in E
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3]

```

In order to have $\#_u(\mathcal{X}, \alpha) \geq 2$ for a given $\alpha \in \overline{\mathbb{Q}(t)}^2$, we shall exclude the solution set of $\text{disc}_u(E[10])$:

```
sat := discrim(E[10], u):
```

Denote by $c_{10}(z_0, z_1) \in \mathbb{Q}(t)[z_0, z_1]$ the coefficients of u^2 in $E[10]$. As $V(E) \cap \{c_{10} = 0\} = \emptyset$, the algorithm deduces from Theorem 4 that $\mathcal{F}_k(u, \mathcal{X})$ is the solution set of the polynomial equations given by E and of the inequation $\text{sat} \neq 0$.

- (3) Turns the inequation $\text{sat} \neq 0$ into the equation $m \cdot \text{sat} - 1 = 0$ by introducing an extra variable m . Finally, it deduces a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, 1)$.

```

H := Groebner[Basis]([op(E), m*sat-1], lexdeg([m, u, z1], [z0, t])):
op(remove(has, H, {m, u, z1}));

```

$$(16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

Thus

$$R = (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1).$$

7.3. SOLVING 3-CONSTELLATIONS USING SECTION 4.3

Denote by $P \in \mathbb{Q}[x, z_0, z_1, t, u]$ the polynomial in 7.1 and by $\mathcal{S}$ the set of polynomials $(P, \partial_x P, \partial_u P, m \cdot tu(u-1) - 1) \subset \mathbb{Q}(t)[m, x, z_0, z_1, u]$. Moreover, recall that $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^4$ is the solution set of the constraints $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u-1) \neq 0$ and that

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{z_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

Goal: Characterize $\mathcal{S}_k(\mathcal{X})$ as the solution set of some polynomial constraints and deduce a nonzero $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, 1)) = 0$.

The algorithm described in Section 4.3:

- (1) Computes a Gröbner basis of the ideal generated by $\mathcal{S}$ for the order $\succ_{\text{bgrelex}}$:

```
G := Groebner[Basis](S, lexdeg([m, x, u, z1], [t, z0])):
```

- (2) Computes the matrix M_{z_1} of the multiplication map $m_{z_1} : f \mapsto z_1 \cdot f$ in the quotient ring $\mathbb{Q}(t, z_0)[m, x, u, z_1]/\langle j(\mathcal{S}) \rangle$, where j is the usual inclusion map of $\mathbb{Q}(t)[m, x, z_0, z_1]$ in $\mathbb{Q}(t, z_0)[m, x, u, z_1]$:

```
L_1, L_2 := Groebner[NormalSet](subs(t = rand(), z0 = rand(), G),
                                tdeg(m, x, u, z1)):
M := Groebner[MultiplicationMatrix](z1, L1, L2, G, tdeg(m, x, u, z1)):
```

- (3) Computes the characteristic polynomial of M_{z_1} and defines χ_{z_1} to be its numerator:

```
xi_z1 := LinearAlgebra[CharacteristicPolynomial](M, T):
chi_z1 := factor( numer(xi_z1) ):
```

- (4) Thus $S_k(\mathcal{X})$ is the solution set of the constraints $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u-1) \neq 0 \wedge \text{disc}_T(\chi_{z_1}) = 0$. An annihilating polynomial of $F(t, 1)$ is given by the discriminant $R \in \mathbb{Q}[t, z_0]$ of χ_{z_1} with respect to T , that is

$$R = 12230590464t^{14} \cdot (16tz_0^2 - 8tz_0 + t - 16)^2 \cdot (1 + (3z_0 - 1)^2t^2 + (-6z_0 - 14)t)^3 \cdot (tz_0 + 1)^{12} \cdot (81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1)^2.$$

7.4. SOLVING 3-CONSTELLATIONS USING SECTION 4.4

Consider (13) and the polynomial $P \in \mathbb{Q}[x, z_0, z_1, t, u]$ of 7.1. Define the set of polynomials $\mathcal{S} := (P, \partial_x P, \partial_u P, m \cdot u(u-1) - 1) \in \mathbb{Q}(t)[m, x, z_0, z_1, u]$.

Goal: Compute a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, 1)) = 0$.

The algorithm described in Section 4.4:

- (1) Applies twice the strategy described in 7.2 to P . It draws at random a prime number p and some $\theta \in \mathbb{F}_p$, say $p = 12301$ and $\theta = 1328$, and applies 7.2 with $\mathcal{S}$ replaced by $(\mathcal{S})|_{t=\theta}$ and $\mathbb{Q}$ replaced by $\mathbb{F}_p$. It outputs

$$7957(z_0 + 11829)(z_0^3 + 4863z_0^2 + 8711z_0 + 3012)(z_0 + 6622)$$

The degree being 5, it sets $b_{z_0} = 5$. Similarly, it obtains $b_t = 3$.

- (2) Computes the 32 first terms of $F(t, 1)$:

$$F_1 = 1 + t + 6t^2 + 54t^3 + 594t^4 + \cdots + 913075994651156584840651326232625946t^{31} \bmod t^{32}$$

- (3) Uses the function `seriestoalgeq` of the Maple package `gfun` [24]:

```
collect(subs(T(t) = z0,
             gfun[seriestoalgeq](series(F1, t, degree(F1, t)), T(t))))[1],
        z0, factor);
```

$$81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1$$

The above polynomial, denoted M , is a candidate for annihilating $F(t, 1)$.

- (4) Proves the guessed polynomial with the below lines

```
series(subs(z0 = F1, M), t, 31);
```

$$0(t^{31})$$

$$\text{Thus } R = 81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1.$$

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